

LESSON 9-3 • HOMEWORK PACKET • SHEET 1 OF 2**Equations from Tables**

Maryland Standard 6.EE.C.9 • Learning target – I can write an equation that matches a table of values.

REMEMBER

- An **equation** like $y = 5x$ is a rule that works for **every** row.
- Same $y \div x$ in every row: $y = (\text{that number}) \times x$. Same $y - x$: $y = x + (\text{that number})$.
- Check your equation with **two** rows.

WORKED EXAMPLEWrite an equation for x : 2, 3, 5 and y : 12, 18, 30.**Step 1** $12 \div 2 = 6$, $18 \div 3 = 6$, $30 \div 5 = 6$ **Step 2** Check: $6 \times 5 = 30$ **Answer** $y = 6x$ **PRACTICE***show your work • label units*

1. Write an equation for each table.

a. x : 1, 2, 3 y : 7, 14, 21 $y =$ _____

b. x : 1, 2, 3 y : 10, 11, 12 $y =$ _____

c. x : 2, 4, 6 y : 10, 20, 30 $y =$ _____

d. x : 5, 6, 7 y : 2, 3, 4 $y =$ _____

e. x : 0, 4, 8 y : 6, 10, 14 $y =$ _____

f. x : 1, 3, 5 y : 9, 27, 45 $y =$ _____

g. x : 2, 3, 4 y : 16, 24, 32 $y =$ _____

h. x : 10, 12, 14 y : 5, 7, 9 $y =$ _____

2. Write the equation. Then complete the table.

x	2	3	4	6	9
y	22	33	44		

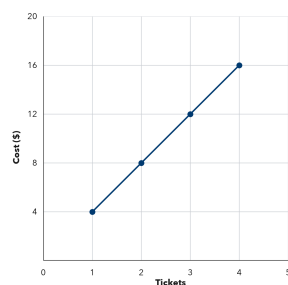
Equation _____

- 3.
- Stretch.**
- A table shows
- x
- : 2, 5, 9 and
- y
- : 0, 3, 7. Write an equation. Then find
- y
- when
- $x = 20$
- .

Answer _____

- 4.
- Tickets.**
- The graph shows the cost of fair tickets. Write an equation for the graph. Then find the cost of 10 tickets.

Answer _____



- 5.
- Find the Error.**
- For
- x
- : 2, 4, 6 and
- y
- : 6, 8, 10, Owen wrote
- $y = 3x$
- because
- $3 \times 2 = 6$
- . Explain his mistake and write the correct equation.

CHECK YOURSELF☐ I can do this on my own☐ I need a little help☐ I have a question about #_____**AT HOME** Make a 3-row table for a rule. Ask someone at home to find your equation.

LESSON 9-3 • HOMEWORK PACKET • SHEET 2 OF 2**Equations from Situations**

Maryland Standard 6.EE.C.9 • Learning target – I can write an equation from a described situation and identify its variables.

REMEMBER

- A **rate** ("each," "per") multiplies x . A **starting amount** or fee is added once.
- $y = (\text{rate}) \times x + (\text{starting amount})$. Check it with one value from the story.

WORKED EXAMPLE

A gym charges \$10 to enter plus \$7 per hour. Write an equation for the cost y of x hours.

Step 1 Rate: 7 per hour, so $7x$. Fee: add 10.

Step 2 Check 3 hours: $7 \times 3 + 10 = 31$

Answer $y = 7x + 10$

PRACTICE

show your work • label units

- Write an equation for each situation.
 - Each pack holds 12 pencils. Pencils y in x packs. $y = \underline{\hspace{2cm}}$
 - Jamal is 3 years older than his sister. Jamal's age y when she is x . $y = \underline{\hspace{2cm}}$
 - A club costs \$20 to join plus \$15 a month. Cost y for x months. $y = \underline{\hspace{2cm}}$
- Bowling.** Shoes cost \$4, plus \$5 per game. Write an equation for the cost y of x games. Find the cost of 4 games.
- Taxi.** A taxi charges \$3 to start plus \$2 per mile. Write an equation for the cost y of x miles. Find the cost of 7 miles.

Answer _____

Answer _____

- Car Wash.** A class has \$30 and earns \$8 per car washed. Complete the table. Write an equation.

Cars (x)	0	1	2	5
Total \$ (y)				

Equation _____

- Find the Error.** For "a \$25 fee plus \$10 an hour," Nia wrote $y = 35x$. Explain her mistake and fix it.

TEST PRACTICE

choose the best answer

- A plumber charges a \$40 fee plus \$25 per hour. Which equation gives the cost c for h hours?

A $c = 40h + 25$ **B** $c = 25h + 40$ **C** $c = 65h$ **D** $c = 25h$
- Which situation matches $y = 3x + 12$?

A 3 cans per box, 12 boxes **B** Start with \$3, earn \$12 per hour

C Start with 12 gal, add 3 gal per min **D** \$12 for 3 people

CHECK YOURSELF

- ☐ I can do this on my own ☐ I need a little help ☐ I have a question about # _____